Breaking and Fixing MacaKey

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Abstract. The sponge construction underpins many modern symmetric primitives, enabling efficient hashing and authenticated encryption. While full-state absorption is known to be secure in keyed sponges, the security of full-state squeezing has remained unclear. Recently, Lefevre and Marhuenda-Beltrán introduced MacaKey, which applies ideas from the summation-truncation hybrid technique of constructing PRFs to the full-state sponge. The authors claimed that MacaKey is provably secure up to the birthday bound in capacity, even when the adversary is allowed to request variable-length outputs. In this work, we revisit this claim and show that MacaKey is insecure as a PRF. We demonstrate a simple four-query distinguishing attack that violates its claimed bound, exploiting the exposure of the full internal state and the resulting loss of secrecy in the capacity portion during squeezing. We then propose a simple modification that restores security with negligible overhead. The modified construction, KeyMacaKey, re-randomizes the internal state after absorption by incorporating a keyed finalization step without requiring an extra permutation call. Further, we show that KeyMacaKey achieves the stronger security of birthday-bound in the full state size than what was claimed for MacaKey.

Keywords: permutation-based cryptography · VIL–VOL PRF · beyond-birthday-bound security · full-state sponge

1 Introduction

The sponge construction, introduced by Bertoni et al. [BDPA07], is a versatile and elegant framework that underlies many modern cryptographic primitives. Its most notable instantiation, *Keccak* [BDPA13], was selected as the winner of the NIST SHA-3 competition, marking a significant shift towards permutation-based cryptography. Beyond hashing, the sponge framework has also found extensive use in the design of authenticated encryption (AE) schemes. A prominent example is Ascon [DEMS21, STMC+25], which not only emerged as a finalist in the CAESAR competition but was later selected as the NIST Lightweight Cryptography standard, further solidifying the sponge paradigm's importance in symmetric-key cryptography.

At a high level, a sponge function operates in two distinct phases: *absorb* and *squeeze*. It maintains an internal state of size b bits, partitioned into two portions—the *outer part* (r bits) and the *inner part* (c bits), such that $b = r + c$. In the canonical sponge operation, the size r of the outer part of the state indicates the *rate of absorption/squeezing*. During the absorbing phase, the input message is first padded and then divided into blocks of r bits, each of which is XORed into the rate portion of the state, followed by an application of the underlying permutation. Once the entire input is absorbed, the function transitions into the squeezing phase, during which the output is generated r bits at a time. The size c of the inner part indicates the *data capacity* of the sponge, i.e., the volume of data the sponge can process without compromising its security. The inner state is always hidden, and provides a security margin against various forms of state recovery and distinguishing attacks. It is well known that for the sponge to remain indifferentiable from a random oracle, r cannot equal b; the security holds only up to $2^{c/2}$ queries [BDPA08].

Keyed-Sponge. While this restriction applies to the *unkeyed* sponge setting, subsequent research explored whether *keyed* constructions could safely operate with larger rates. Mennink et al. [MRV15] demonstrated that in the absorption phase of a keyed sponge, it is indeed possible to set the rate r equal to the full state size b, since the presence of a secret key alters the adversarial model. However, for the squeezing phase, the restriction persists—it has long been believed that squeezing b bits at a time remains insecure, as it would violate the assumptions that ensure pseudorandomness of the output. Some constructions, such as Muffler [BBN22] and Kirby [LBD23], permit both full-length absorption and squeezing, but their output is inherently limited to at most b-bits.

MacaKey. Recently, Lefevre and Marhuenda-Beltrán [LB25a] revisited this question in their work introducing the MacaKey construction, a permutation-based *variable input length* and *variable output length* keyed design. They claimed that it is possible not only to absorb over the entire state but also to squeeze the full b-bit output while retaining provable security. As the rate r directly governs the efficiency of a sponge-based construction, such a result represents a significant advancement in optimizing the performance of permutation-based designs. The authors also presented a formal security bound for MacaKey in the random permutation model and extended their analysis to the multi-user setting, suggesting strong practical implications for the full-state keyed sponge paradigm.

1.1 Our Contributions

In this work, we make two primary contributions: (1) we identify a critical flaw in the security proof of MacaKey and show that MacaKey is, in fact, an insecure PRF by exhibiting a distinguishing attack; (2) we patch the construction to propose KeyMacaKey, and show that it achieves an optimal PRF security of birthday-bound in the full state size.

Insecurity of the MacaKey construction. Our first contribution concerns the security of the MacaKey construction proposed by Lefevre and Marhuenda-Beltrán [LB25a]. We carefully reanalyze their proof and identify a critical flaw in the argument establishing MacaKey's security. In fact, we go further to show that this flaw is not merely an oversight in the proof, but that the construction itself is insecure. We present a simple 4-query distinguishing attack that breaks the claimed security bound of MacaKey.

At a high level, the vulnerability stems from the fact that, in MacaKey, the adversary obtains control over the entire b-bit state. Consequently, at the beginning of the squeezing phase, the internal c-bit portion of the state (which in a standard sponge would remain secret) becomes exposed to adversarial influence. This loss of secrecy allows the adversary to mount a distinguishing attack that invalidates the claimed security guarantees.

The modified construction KeyMacaKey. As our second contribution, we present a simple yet effective modification to the MacaKey design to restore its security. We observe that blinding the internal state with the key at the end of the absorption phase re-randomizes the internal state, thereby ensuring that the entire state remains secret at the beginning of the squeezing phase. We refer to this modified construction as KeyMacaKey.

We provide a formal security analysis of KeyMacaKey, establishing its generic security in the random permutation model. The key-blinding step allows us to prove a significantly stronger security guarantee than that claimed by Lefevre and Marhuenda-Beltrán. Our analysis first considers the single-user setting, yielding a security bound of the order

Table 1: Comparison of full state PRF Constructions

Design	Security	Output Size	# Permutations	Reference
Farfalle	No provable claim	variable	1	[BDH+17]
Farasha	BB (state, key)	variable	1	[ABJ+22]
Kirby	BBB	1 block	1	[LBD23]
Muffler	BBB	1 block	1	[BBN22]
MacaKey v1	Not secure	variable	1	[LB25a]
MacaKey v2	BBB	variable	2	[LB25b]
KeyMacaKey	BBB	variable	1	This work

$\min\{T/2^c, D^2/2^b, T/2^\kappa\}$. Here D and T represent the data complexity (the number of construction queries) and time complexity (the number of permutation queries) respectively, and κ represents the key-size. We then extend our proof to the multi-user setting, where the bound incurs an additional term of $(\mu T)/2^\kappa$, μ being the number of users. This is a substantial improvement over the $D^2/2^c + LT/2^c$ term reported by Lefevre and Marhuenda-Beltrán, where L is a parameter that can be potentially as high as D. Overall, our results demonstrate that KeyMacaKey achieves stronger security guarantees comparable to traditional keyed sponge constructions, while preserving the efficiency benefits of full-state operation.

1.2 Related Works

Permutation-based variable-input-length to variable-output-length (VIL–VOL) pseudo-random functions have been studied extensively, particularly in the context of achieving beyond-birthday-bound (BBB) security under standard assumptions.

MacaKey. The MacaKey construction of Lefevre and Marhuenda Beltrán [LB25a] proposes a VIL–VOL PRF based on the full-state sponge.

The original version (v1), as presented in the initial ePrint submission of [LB25a], employs a single permutation everywhere and claims standard birthday-bound security. Our work analyses this variant and shows that the corresponding proof does not support the claimed bound; in particular, we give a concrete distinguishing attack demonstrating a violation of the stated security guarantee.

A subsequent revision (v2) [LB25b], released after the submission of our work, acknowledges our work, and modifies the design by employing two independent permutations to separate key absorption from squeezing. This variant avoids the specific attack and proof issue identified for v1 and achieves a security level comparable to that of our construction, which is beyond the birthday-bound in capacity or key-size.

Other full-state sponge-based VIL-VOL PRFs. To the best of our knowledge, KeyMacaKey is the first construction to explicitly prove beyond-birthday-bound security for a full-state sponge-based VIL-VOL PRF. There also exist permutation-based constructions such as Farfalle [BDH+17] and its provably secure variant Farasha [ABJ+22] that provide VIL-VOL functionality; however, their security is limited to the birthday bound in the minimum of the permutation size and the key size. In contrast, constructions such as Kirby [LBD23] and Muffler [BBN22] achieve beyond-birthday-bound security, but restrict the output length to a single block. A summary of these comparisons is provided in Table 1.

2 Preliminaries

For all $a \leq b \in \mathbb{N}$, let $[b]$ and $[a,b]$ denote the sets $\{1,2,\ldots,b\}$ and $\{a,a+1,\ldots,b\}$ respectively. Let $\{0,1\}^n$ denote the set of bit strings of length n, and $\{0,1\}^*$ denote the set of bit strings of arbitrary length. For any bit string x of length at least n bits, let $\lceil x \rceil_n$ (resp. $\lfloor x \rfloor_n$) denote the most (resp. least) significant n bits of x.

For a finite set $\mathcal{X}$, $\mathsf{X} \xleftarrow{\$} \mathcal{X}$ denotes the uniform and random sampling of X from $\mathcal{X}$, and $\mathsf{X} \xleftarrow{\mathsf{wor}} \mathcal{X}$ denotes without replacement sampling of X from $\mathcal{X}$. Let $\mathsf{Perm}(b)$ denote the set of all permutations over b-bits. Given any $M \in \{0,1\}^{b*m}$, let $(M_1,\ldots,M_m) \xleftarrow{b} M$ denote the parsing of M in m blocks such that $M := M_1\|M_2\|...\|M_m$. Let $\mathsf{pad} : \{0,1\}^* \to \{0,1\}^{b*}$ denote an injective function such that for all $(M_1,\ldots,M_m) \xleftarrow{b} \mathsf{pad}(M)$, $M_m \neq 0^b$.

2.1 Expected Maximum Multicollision in a Uniform Random Sample

In this section we revisit some results on multicollisions as discussed in [CJN20]. In their notation, let $\mathsf{X}_1,\ldots,\mathsf{X}_q \xleftarrow{\$} \mathcal{D}$ where $|\mathcal{D}| = N$. Denote the maximum multicollision random variable for the sample as $\mathsf{mc}_{q,N}$ and write $\mathsf{mcoll}(q,N)$ to denote $\mathbb{E}[\mathsf{mc}_{q,N}]$.

Proposition 1. *[CJN20] For $N \geq 4$, $n = \log_2 N$,*

$$\mathsf{mcoll}(q,N) \leq \begin{cases} \frac{4\log_2 q}{\log_2 \log_2 q} & \text{if } 4 \leq q \leq N \\ 5n\left\lceil \frac{q}{nN} \right\rceil & \text{if } N < q \end{cases}$$

Remark 1. [CJN20] Similar bounds as in proposition 1 can be achieved in the case of non-uniform samplings such as $\mathsf{X}_1,\ldots,\mathsf{X}_q \xleftarrow{\mathsf{wor}} \{0,1\}^b$.

2.2 Pseudorandom Functions: Definition and Security Model

For a keyed function $\mathsf{F} : \mathcal{K} \times \mathcal{D} \to \mathcal{R}$ (where $\mathcal{K} = \{0,1\}^\kappa, \kappa \in \mathbb{N}$ is the key space), we define the PRF-advantage of an adversary $\mathscr{A}$ against F as

$$\mathbf{Adv}^{\mathsf{prf}}_{\mathsf{F}}(\mathscr{A}) := \left| \Pr_{\mathsf{K} \xleftarrow{\$} \mathcal{K}}\left[\mathscr{A}^{\mathsf{F}_\mathsf{K}} = 1\right] - \Pr_{f \xleftarrow{\$} \mathsf{Func}(\mathcal{D},\mathcal{R})}\left[\mathscr{A}^f = 1\right] \right|.$$

Here, $\mathscr{A}^{\mathsf{F}_\mathsf{K}}$ and $\mathscr{A}^f$ denote $\mathscr{A}$'s response after its interaction with F_K and f respectively.

PRFs in the random permutation model. For our analysis, we consider keyed constructions $\mathsf{C}^P : \mathcal{K} \times \mathcal{D} \to \mathcal{R}$ which make queries to a randomly permutation $P \xleftarrow{\$} \mathsf{Perm}(b)$ which can be evaluated by the adversary $\mathscr{A}$. We define the PRF-advantage of $\mathscr{A}$ against C^P as

$$\mathbf{Adv}^{\mathsf{prf}}_{\mathsf{C}^P}(\mathscr{A}) := \left| \Pr_{\substack{\mathsf{K} \xleftarrow{\$} \mathcal{K} \\ P \xleftarrow{\$} \mathsf{Perm}(b)}}\left[\mathscr{A}^{\mathsf{C}^P_\mathsf{K},P} = 1\right] - \Pr_{\substack{R \xleftarrow{\$} \mathsf{Func}(\mathcal{D},\mathcal{R}) \\ P \xleftarrow{\$} \mathsf{Perm}(b)}}\left[\mathscr{A}^{R,P} = 1\right] \right|.$$

Here, $\mathscr{A}^{\mathsf{C}^P_\mathsf{K},P}$ denotes $\mathscr{A}$'s response after its interactions with C^P_K and P respectively. Similarly, $\mathscr{A}^{R,P}$ denotes $\mathscr{A}$'s response after its interactions with R and P respectively. We call $\mathscr{A}$'s queries to its first oracle (C^P_K or R) as *construction queries* and its queries to the second oracle (P) as *primitive queries*.

2.3 H-coefficient Technique

Consider an adversary $\mathscr{A}$, which is deterministic and computationally unbounded, attempting to distinguish between the real oracle, denoted as $\mathcal{O}_{\text{re}}$, and the ideal oracle, denoted as $\mathcal{O}_{\text{id}}$. The interaction of $\mathscr{A}$ with its oracle is captured by the query-response tuple denoted as ω. In certain scenarios, after the query-response phase of the game, the oracle may choose to reveal additional information to the distinguisher. In such cases, the extended definition of the transcript may include that additional information. Let τ_{re} (respectively, τ_{id}) represent the random transcript variable when $\mathscr{A}$ interacts with $\mathcal{O}_{\text{re}}$ (respectively, $\mathcal{O}_{\text{id}}$). The probability of realizing a specific transcript ω in the security game with an oracle $\mathcal{O}$ is referred to as the *interpolation probability* of ω with respect to $\mathcal{O}$. Given the determinism of $\mathscr{A}$, this probability depends solely on the oracle $\mathcal{O}$ and the transcript ω. A transcript ω is considered *realizable* if $\Pr[\tau_{\text{id}} = \omega] > 0$. In this paper, $\mathcal{O}_{\text{re}} = (\mathsf{PRF}_{\mathsf{K}}, P)$, $\mathcal{O}_{\text{id}} = (\Gamma, P)$, and the adversary aims to distinguish $\mathcal{O}_{\text{re}}$ from $\mathcal{O}_{\text{id}}$.

Proposition 2 (H-coefficient technique [Pat91, Pat08])**.** *Let Ω be the set of all realizable transcripts. For some $\epsilon_{\mathsf{bad}}, \epsilon_{\mathsf{ratio}} > 0$, suppose there is a set $\Omega_{\mathsf{bad}} \subseteq \Omega$ satisfying the following:*

- $\Pr[\tau_{\text{id}} \in \Omega_{\mathsf{bad}}] \leq \epsilon_{\mathsf{bad}}$;
- *For any $\omega \notin \Omega_{\mathsf{bad}}$,*
$$\frac{\Pr[\tau_{\text{re}} = \omega]}{\Pr[\tau_{\text{id}} = \omega]} \geq 1 - \epsilon_{\mathsf{ratio}}.$$

Then for any adversary $\mathscr{A}$, we have the following bound on its distinguishing advantage:

$$\mathbf{Adv}^{\text{dist}}_{\mathcal{O}_{\text{re}},\mathcal{O}_{\text{id}}}(\mathscr{A}) \leq \epsilon_{\mathsf{bad}} + \epsilon_{\mathsf{ratio}}.$$

A proof of Proposition 2 can be found in multiple papers including [Pat08, CS14, MN17].

3 Breaking the MacaKey Construction

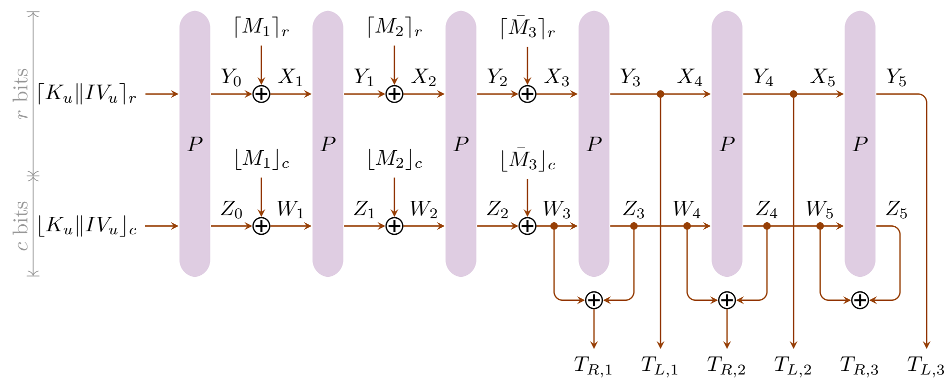

Figure 1: The MacaKey construction for a 3-block input and 3-block output. $\bar{M}_3$ is the last block of M padded to b bits. The final output is $\lceil T_L \| T_R \rceil_n$ for a requested output length n satisfying $2b < n \leq 3b$, where $T_L := T_{L,1} \| T_{L,2} \| T_{L,3}$ and $T_R := T_{R,1} \| T_{R,2} \| T_{R,3}$.

The MacaKey construction, proposed by Lefevre and Marhuenda-Beltrán [LB25a], is a permutation-based variable-input-length and variable-output-length PRF. It is based on the keyed sponge with full-state absorption, with the additional observation that full-state

Algorithm 1 The MacaKey construction based on the permutation P

Require: input $(u, M, n) \in [\mu] \times \{0,1\}^* \times \mathbb{N}^+$
Ensure: output $T \in \{0,1\}^n$

$\mathsf{tmp} \leftarrow P(K_u \| IV_u)$
$Y_0 \leftarrow \lceil \mathsf{tmp} \rceil_r;\ Z_0 \leftarrow \lfloor \mathsf{tmp} \rfloor_c$
$m \leftarrow \lceil |M|/b \rceil$
$(M_1, \ldots, M_m) \xleftarrow{b} \mathsf{pad}(M)$
for $j = 1$ **to** $m-1$ **do**
  $X_j \leftarrow Y_{j-1} \oplus \lceil M_j \rceil_r;\ W_j \leftarrow Z_{j-1} \oplus \lceil M_j \rfloor_c$
  $\mathsf{tmp} \leftarrow P(X_j \| W_j)$
  $Y_j \leftarrow \lceil \mathsf{tmp} \rceil_r;\ Z_j \leftarrow \lfloor \mathsf{tmp} \rfloor_c$
$X_m \leftarrow Y_{m-1} \oplus \lceil M_m \rceil_r;\ W_m \leftarrow Z_{m-1} \oplus \lceil M_m \rfloor_c$
$\mathsf{tmp} \leftarrow P(X_m \| W_m)$
$Y_m \leftarrow \lceil \mathsf{tmp} \rceil_r;\ Z_m \leftarrow \lfloor \mathsf{tmp} \rfloor_c$
$T_L \leftarrow Y_m;\ T_R \leftarrow W_m \oplus Z_m$
$l \leftarrow \lceil n/b \rceil$
for $j = m+1$ **to** $m+l$ **do**
  $X_j \leftarrow Y_{j-1};\ W_j \leftarrow Z_{j-1}$
  $\mathsf{tmp} \leftarrow P(X_j \| W_j)$
  $Y_j \leftarrow \lceil \mathsf{tmp} \rceil_r;\ Z_j \leftarrow \lfloor \mathsf{tmp} \rfloor_c$
  $T_L \leftarrow T_L \| Y_j;\ T_R \leftarrow T_R \| (W_j \oplus Z_j)$
$T \leftarrow T_L \| T_R$
return $\lceil T \rceil_n$

squeezing can also be achieved through a feed-forward of the capacity part of the state. Algorithm 1 describes the algorithm of MacaKey, and Figure 1 illustrates the execution flow of MacaKey when both input and output lengths are between $2b$ and $3b$.

The authors prove that MacaKey is a secure PRF up to $O(DT/2^c)$ queries, which is the birthday bound in capacity. The security proof of MacaKey, as presented in [LB25a], proceeds via a standard H-coefficient argument, in which the real and ideal worlds are shown to remain indistinguishable except when certain "bad" transcript events occur. In this framework, bounding the probability of the bad events is central to establishing pseudorandomness. In particular, the analysis treats collisions in the inner component of permutation inputs as occurring only with birthday-type probability, reflecting the intuition that this portion of the state remains hidden from the adversary despite full-state absorption and squeezing.

We briefly recall the bad event most relevant to our analysis before identifying a gap in the argument.

3.1 Flaw in the Proof

The authors define a bad event which, in our terminology, can be expressed as

$$\texttt{IColl} : \exists (i,j) \neq (i',j') \text{ s.t. } W_j^i = W_{j'}^{i'},$$

However, in their analysis of the probability of `IColl`, the authors overlook an important case: when $i \neq i'$ and $j = l_i, j' = l_{i'}$. In this scenario, the adversary can issue two queries that differ only in the last block of the padded message, where the difference lies solely in the upper r bits. As a result, the equality $W_{l_i}^i = W_{l_{i'}}^{i'}$ still holds, leading to an unaccounted collision event.

3.2 A Four Message Attack

We now leverage the flaw identified in the proof to construct an adversary that makes only four PRF queries to distinguish $\mathsf{MacaKey}$ from a random function.

- The adversary selects two distinct messages $M^1 \neq M^2$ such that $|\mathsf{pad}(M^1)| = |\mathsf{pad}(M^2)| = b$ and $\lfloor \mathsf{pad}(M^1) \rfloor_c = \lfloor \mathsf{pad}(M^2) \rfloor_c$.

- It then makes two PRF queries (M^1, b) and (M^2, b), obtaining the corresponding outputs $T_L^1 \| T_R^1$ and $T_L^2 \| T_R^2$, respectively.

- Next, the adversary prepares two additional messages (M^3, b) and (M^4, b), where

$$\mathsf{pad}(M^3) := \mathsf{pad}(M^1) \| T_L^2 \| M', \quad \mathsf{pad}(M^4) := \mathsf{pad}(M^2) \| T_L^1 \| (M' \oplus T_R^1 \oplus T_R^2),$$

  for some $M' \in \{0,1\}^c$.

- Finally, the adversary outputs 1 if $\mathsf{MacaKey}(K, IV, M^3, b) = \mathsf{MacaKey}(K, IV, M^4, b)$, and 0 otherwise.

Analysis. It is easy to observe that $W_1^1 = W_1^2 = W_1^3 = W_1^4$. Further, $Z_1^1 \oplus Z_1^2 = T_R^1 \oplus T_R^2$. Hence,

$$X_2^3 = T_L^2 \oplus T_L^1 = X_2^4;$$

$$W_2^3 = Z_1^1 \oplus M' = Z_1^2 \oplus T_R^1 \oplus T_R^2 \oplus M' = W_2^4.$$

Thus, when interacting with the real oracle, the adversary outputs 1 with probability 1. When interacting with the ideal oracle, it only outputs 1 with a probability of about $1/2^b$. Thus the attack advantage is about $1 - 1/2^b$.

3.3 Attack on MacaKey with Full State XOR

Our attack on $\mathsf{MacaKey}$ relies on the adversary's ability to select messages M^1 and M^2 such that $\lfloor M^1 \rfloor_c = \lfloor M^2 \rfloor_c$. However, this attack no longer applies if we set $c = b$ during the squeezing phase (i.e., the XOR is applied to the full state while squeezing). Moreover, the security bound for $\mathsf{MacaKey}$ in [LB25a] does not include any r or 2^r term in the denominator, which might initially suggest that choosing $c = b$ resolves the issue. Nevertheless, as we demonstrate below, this is not the case.

- The adversary selects two distinct messages $M^1 \neq M^2$ such that $|\mathsf{pad}(M^1)| = |\mathsf{pad}(M^2)| = b$.

- It then makes two PRF queries (M^1, b) and (M^2, b), obtaining the corresponding outputs T^1 and T^2, respectively.

- Next, the adversary prepares two additional messages (M^3, b) and (M^4, b), where

$$\mathsf{pad}(M^3) := \mathsf{pad}(M^1) \| M', \quad \mathsf{pad}(M^4) := \mathsf{pad}(M^2) \| (M' \oplus M^1 \oplus M^2 \oplus T^1 \oplus T^2),$$

  for some $M' \in \{0,1\}^b$.

- Finally, the adversary outputs 1 if $\mathsf{MacaKey}(K, IV, M^3, b) = \mathsf{MacaKey}(K, IV, M^4, b)$, and 0 otherwise.

Analysis. It is easy to observe that $Z_1^1 \oplus Z_1^2 = M^1 \oplus M^2 \oplus T^1 \oplus T^2$. Hence,

$$W_2^4 = M' \oplus M^1 \oplus M^2 \oplus T^1 \oplus T^2 \oplus Z_1^2 = M' \oplus Z_1^1 = W_2^3.$$

Here, W_i represent the full inputs to the permutation. As before, when interacting with the real oracle, the adversary outputs 1 with probability 1, and when interacting with the ideal oracle, it only outputs 1 with a probability of about $1/2^b$, so the attack advantage is about $1 - 1/2^b$.

4 The KeyMacaKey Construction

Algorithm 2 The KeyMacaKey construction based on an extra key addition to MacaKey

Require: input $(u, M, n) \in [\mu] \times \{0,1\}^* \times \mathbb{N}^+$
Ensure: output $T \in \{0,1\}^n$

$\mathsf{tmp} \leftarrow P(K_u \| IV_u)$
$Y_0 \leftarrow \lceil \mathsf{tmp} \rceil_r$; $Z_0 \leftarrow \lfloor \mathsf{tmp} \rfloor_c$
$m \leftarrow \lceil |M|/b \rceil$
$(M_1, \ldots, M_m) \xleftarrow{b} \mathsf{pad}(M)$
for $j = 1$ **to** $m - 1$ **do**
  $X_j \leftarrow Y_{j-1} \oplus \lceil M_j \rceil_r$; $W_j \leftarrow Z_{j-1} \oplus \lceil M_j \rfloor_c$
  $\mathsf{tmp} \leftarrow P(X_j \| W_j)$
  $Y_j \leftarrow \lceil \mathsf{tmp} \rceil_r$; $Z_j \leftarrow \lfloor \mathsf{tmp} \rfloor_c$
$X_m \leftarrow Y_{m-1} \oplus \lceil M_m \oplus 0^{b-\kappa} \| K_u \rceil_r$; $W_m \leftarrow Z_{m-1} \oplus \lceil M_m \oplus 0^{b-\kappa} \| K_u \rfloor_c$
$\mathsf{tmp} \leftarrow P(X_m \| W_m)$
$Y_m \leftarrow \lceil \mathsf{tmp} \rceil_r$; $Z_m \leftarrow \lfloor \mathsf{tmp} \rfloor_c$
$l \leftarrow \lceil n/b \rceil$
$T \leftarrow (W_m \oplus Z_m) \| Y_m$
for $j = m + 1$ **to** $m + l$ **do**
  $X_j \leftarrow Y_{j-1}$; $W_j \leftarrow Z_{j-1}$
  $\mathsf{tmp} \leftarrow P(X_j \| W_j)$
  $Y_j \leftarrow \lceil \mathsf{tmp} \rceil_r$; $Z_j \leftarrow \lfloor \mathsf{tmp} \rfloor_c$
  $T \leftarrow T \| (W_j \oplus Z_j) \| Y_j$
return $\lceil T \rceil_n$

We show that the security of MacaKey can be restored by introducing a simple yet effective modification, inspired by the keyed finalization of Ascon. More precisely, we add $0^{b-\kappa} \| K$ to the last padded message block. We call the modified construction KeyMacaKey. Algorithm 2 describes the algorithm of KeyMacaKey, and Figure 2 illustrates the execution flow of KeyMacaKey when both input and output lengths are between $2b$ and $3b$.

We prove the following theorems concerning the single-user and multi-user security of KeyMacaKey.

Theorem 1. *For any* PRF *adversary making q_p primitive queries and q_c construction queries with a total number of construction query blocks σ, and total number of output digest blocks Λ,*

$$\begin{aligned}\mathbf{Adv}^{\mathsf{PRF}}_{\mathsf{KeyMacaKey}}(q_p, q_c, \sigma, \Lambda) \leq\ & \frac{2q_c + q_p}{2^\kappa} + \frac{q_p(2\sigma + 6\Lambda + q_c)}{2^b} \\ & + \frac{\sigma(2\sigma + 2q_c + 2\Lambda + 1) + \Lambda(\Lambda + 2q_c + 1) + q_c(q_c + 1)}{2^b} \\ & + \frac{\sigma\Lambda}{2^{\min\{b, r+\kappa\}}} + \frac{\mathsf{mcoll}(\Lambda, 2^c) q_c}{2^r} + \frac{2\mathsf{mcoll}(\Lambda, 2^r) q_p}{2^c}.\end{aligned}$$

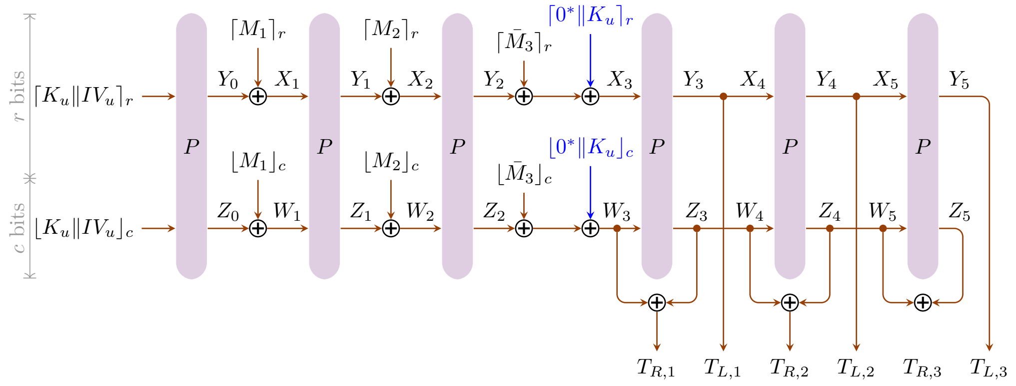

Figure 2: The KeyMacaKey construction for a 3-block input and 3-block output. $\bar{M}_3$ is the last block of M padded to b bits. The final output is $\lceil T_{R,1} \| T_{L,1} \| T_{R,2} \| T_{L,2} \| T_{R,3} \| T_{L,3} \rceil_n$ for a requested output length n satisfying $2b < n \leq 3b$.

Theorem 2. *Given μ users, let $\vec{IV}$ and $\vec{K}$ denote the vector of all IVs and keys used by the users. Let $\mathcal{IV}$ denote the set of all distinct IVs and $\nu^{IV} = |\{u \in [\mu] \mid \vec{IV}[u] = IV\}|$. Let $\mathcal{L} = \max_{IV \in \mathcal{IV}} u^{IV}$. For any* PRF *adversary making q_p primitive queries and q_c construction queries to the users, with a total number of construction query blocks σ, and total number of output digest blocks Λ,*

$$\begin{aligned}
\mathbf{Adv}^{\mathsf{PRF}}_{\mathsf{KeyMacaKey}}(q_p, q_c, \sigma, \Lambda, \vec{IV}, \vec{K}) \leq\ & \frac{2q_c + \mathcal{L}q_p + \mathcal{L}\mu}{2^\kappa} + \frac{q_p(2\sigma + 6\Lambda + q_c)}{2^b} \\
&+ \frac{\sigma(2\sigma + 2q_c + 2\Lambda + 1) + \Lambda(\Lambda + 2q_c + 1) + q_c(q_c + \mu)}{2^b} \\
&+ \frac{\sigma\Lambda}{2^{\min\{b, r+\kappa\}}} + \frac{\mathsf{mcoll}(\Lambda, 2^c)q_c}{2^r} + \frac{2\mathsf{mcoll}(\Lambda, 2^r)q_p}{2^c}.
\end{aligned}$$

Interpretation of Theorems 1 and 2. For the purpose of interpreting the theorem, we identify q_c, σ, and Λ with the data complexity D, and q_p with the time complexity T, under the standard assumption that $D \leq T$. We further assume $r, c \geq 128$, so that for $D \leq 2^{128}$, $\mathsf{mcoll}(\Lambda, 2^r)$ and $\mathsf{mcoll}(\Lambda, 2^c)$ are both constants.

Under these assumptions, the dominating terms in the bound for Theorem 1 (up to small constant factors) are

$$\frac{DT}{2^b} + \frac{T}{2^c} + \frac{D}{2^r} + \frac{T}{2^\kappa},$$

which is close to optimal for a sponge-based mode. For instance, using the 320-bit Ascon permutation with $r \approx 160$ and $c \approx 160$, the adversary advantage remains negligible even after absorbing/squeezing 2^{128} data blocks of size 320 bits each.

In case of Theorem 2, i.e., in the multi-user setting, the only modification is that the term $T/2^\kappa$ is replaced by the term $\mu T/2^\kappa$, which corresponds to the standard multiuser degradation.

Optimal choices of r and c. We note that for this construction, the notions of rate and capacity do not apply in the traditional sense, since KeyMacaKey is a full-state PRF. The parameter c instead behaves as a memory overhead, and choosing $c = \kappa$ is sufficient from a security standpoint.

5 Proofs of Theorem 1 and 2

We begin by establishing Theorem 1, which proves the security of KeyMacaKey in the single-user setting. This result is then extended to the multi-user setting in Theorem 2, presented in Section 5.4.

5.1 Description of the Real World

In the real world, the key K is sampled randomly as $K \xleftarrow{\$} \{0,1\}^\kappa$. Both the construction oracle and the adversary has access to a stateful primitive oracle, which simulates a random permutation P through lazy sampling. The construction oracle answers its own queries following the KeyMacaKey construction as defined above and accessing P through the primitive oracle. After all the queries have been answered, the adversary's transcript is of the form

$$\tau_{\text{re,on}} := \left((M^i, n_i, T^i)_{i\in[q_c]}; (U_i, V_i, \mathsf{dir}_i)_{i\in[q_p]}\right),$$

where $\mathsf{dir}_i = +$ (resp. $\mathsf{dir}_i = -$) denotes that the i-th primitive query from the adversary was a forward query with U^i (resp. inverse query with V^i).

At this point, P is a partial permutation, which we write as $P = P_\star \cup P_\dagger$, where $P_\star := (U_i, V_i)_{i\in[q_p]}$ contains the query-response pairs for the adversary's primitive queries, while $P_\dagger$ contains the query-response pairs corresponding to the queries from the construction oracle. Before the adversary's final output, $P_\dagger$ is also revealed to the adversary. (Note that the key K can be determined from $P_\dagger$). The adversary's extended transcript now becomes

$$\tau_{\text{re}} := \left((M^i, n_i, T^i_L \| T^i_R)_{i\in[q_c]}; (U_i, V_i, \mathsf{dir}_i)_{i\in[q_p]}; P_\dagger; K\right).$$

Denoting $\gamma := (M^i, n_i, T^i)_{i\in[q_c]}$ and $\delta := (\mathsf{dir}_i)_{i\in[q_p]}$, we can rewrite this as $\tau_{\text{re}} = (\gamma, \delta, P, K)$. We observe that the stochastic part of τ_{re} is entirely contained in (P, K). Thus, for any realizable transcript $\omega = (\gamma^\omega, \delta^\omega, P^\omega, K^\omega)$,

$$\Pr[\tau_{\text{re}} = \omega] = \Pr[P = P^\omega, K = K^\omega] = \frac{1}{(2^b)_{|P|}} \cdot \frac{1}{2^\kappa}. \tag{1}$$

5.2 Description of the Ideal World

For simplicity, we use the same notation for the transcripts in both the real world and the ideal world. The adversary has access to a stateful primitive oracle, which simulates a random permutation P through lazy sampling. The construction oracle, on each query M^i, n_i, samples and returns $T^i \xleftarrow{\$} \{0,1\}^{n_i}$. Using notation as before, the adversary's transcript at the end of the query phase is

$$\tau_{\text{id,on}} := \left((M^i, n_i, T^i)_{i\in[q_c]}; (U_i, V_i, \mathsf{dir}_i)_{i\in[q_p]}\right).$$

Before the adversary's final output, $(P_\dagger, K)$ is sampled as in Algorithm 3 and returned to the adversary. The adversary's extended transcript now becomes

$$\tau_{\text{id}} := (\gamma, \delta, P, K).$$

We next define six bad events that can occur during the sampling of $(P_\dagger, K)$.

`BAD1 :` The Abort instruction on line 15 in Algorithm 3 is executed.

`BAD2 :` The Abort instruction on line 19 in Algorithm 3 is executed.

`BAD3 :` The Abort instruction on line 30 in Algorithm 3 is executed.

Algorithm 3 Sampling of $(P_\dagger, K)$ in the Ideal World in the Proof of Theorem 1.

$P_\dagger \leftarrow \emptyset;\ \mathcal{S}_{\text{pfx}} \leftarrow \emptyset$
$K \xleftarrow{\$} \{0,1\}^\kappa$
$Y_0 \xleftarrow{\$} \{0,1\}^r;\ Z_0 \xleftarrow{\$} \{0,1\}^c$
$P_\dagger(K \| IV) \leftarrow Y_0 \| Z_0$
for $i = 1$ to q_c **do**
$m_i \leftarrow \lceil |M^i|/b \rceil$
$(M_1^i, \ldots, M_{m_i}^i) \xleftarrow{b} \mathsf{pad}(M^i)$
$Y_0^i \leftarrow Y_0;\ Z_0^i \leftarrow Z_0$
for $j = 1$ to $m_i - 1$ **do**
$X_j^i \leftarrow Y_{j-1}^i \oplus \lceil M_j^i \rceil_r;\ W_j^i \leftarrow Z_{j-1}^i \oplus \lfloor M_j^i \rfloor_c$
if $M_1^i \| \ldots \| M_j^i \in \mathcal{S}_{\text{pfx}}$ **then**
$\mathsf{tmp} \leftarrow P_\dagger(X_j^i \| W_j^i)$
$Y_j^i \leftarrow \lceil \mathsf{tmp} \rceil_r;\ Z_j^i \leftarrow \lfloor \mathsf{tmp} \rfloor_c$
else if $X_j^i \| W_j^i \in \mathsf{domain}(P_\dagger)$ **then**
Abort
else
$Y_j^i \xleftarrow{\$} \{0,1\}^r;\ Z_j^i \xleftarrow{\$} \{0,1\}^c$
if $Y_j^i \| Z_j^i \in \mathsf{range}(P_\dagger)$ **then**
Abort
$P_\dagger(X_j^i \| W_j^i) \leftarrow Y_j^i \| Z_j^i$
$\mathcal{S}_{\text{pfx}} \leftarrow \mathcal{S}_{\text{pfx}} \cup \{M_1^i \| \ldots \| M_j^i\}$
$X_{m_i}^i \leftarrow Y_{m_i-1}^i \oplus \lceil M_{m_i}^i \oplus 0^{b-\kappa} \| K \rceil_r;\ W_{m_i}^i \leftarrow Z_{m_i-1}^i \oplus \lfloor M_{m_i}^i \oplus 0^{b-\kappa} \| K \rfloor_c$
$l_i \leftarrow \lceil n_i/b \rceil$
$\tilde{T}^i \xleftarrow{\$} \{0,1\}^{bl_i - n_i}$
$(T_1^i, \ldots, T_{l_i}^i) \xleftarrow{b} T^i \| \tilde{T}^i$
for $j = m_i$ to $m_i + l_i - 1$ **do**
$T_{R,j-m_i+1}^i \leftarrow \lceil T_{j-m_i+1}^i \rceil_c$
$T_{L,j-m_i+1}^i \leftarrow \lfloor T_{j-m_i+1}^i \rfloor_r$
if $X_j^i \| W_j^i \in \mathsf{domain}(P_\dagger)$ **then**
Abort
$Y_j^i \leftarrow T_{L,j-m_i+1}^i;\ Z_j^i \leftarrow W_j^i \oplus T_{R,j-m_i+1}^i$
if $Y_j^i \| Z_j^i \in \mathsf{range}(P_\dagger)$ **then**
Abort
$P_\dagger(X_j^i \| W_j^i) \leftarrow Y_j^i \| Z_j^i$
if $j < m_i + l_i - 1$ **then**
$X_{j+1}^i \| W_{j+1}^i \leftarrow Y_j^i \| Z_j^i$
return $P_\dagger$

BAD4 : The Abort instruction on line 33 in Algorithm 3 is executed.

BAD5 : $\mathsf{domain}(P_\dagger) \cap \mathsf{domain}(P_\star) \neq \emptyset$.

BAD6 : $\mathsf{range}(P_\dagger) \cap \mathsf{range}(P_\star) \neq \emptyset$.

Define $\mathtt{BAD} := \bigcup_{i=1}^{6} \mathtt{BADi}$. We say that a transcript τ is good if BAD does not occur, i.e., if Algorithm 3 terminates without aborting and $P := P_\star \cup P_\dagger$ is a partial permutation. As before, (P, K) determines the entire transcript. Consider a good transcript (P^ω, K^ω) with $P^\omega = P_\star^\omega \cup P_\dagger^\omega$. The sampling of $P_\star$ is done using the lazy sampling of an ideal permutation, so

$$\Pr[P_\star = P_\star^\omega] = \frac{1}{(2^b)_{q_p}}.$$

For sampling $P_\dagger$, in the online phase, $\sum_{i=1}^{q_c} n_i$ bits are first sampled during the online phase. Then, in the offline sampling phase, $b\Lambda - \sum_{i=1}^{q_c} n_i$ bits are sampled for the squeezing calls (on line 23 of Algorithm 3) and $b(1 + |\mathcal{S}_{\mathrm{pfx}}|)$ bits are sampled for the absorbing calls (on lines 3 and 16 of Algorithm 3), to make a total of $b(1 + |\mathcal{S}_{\mathrm{pfx}}| + \Lambda)$ bits over the two phases. Thus,

$$\Pr[P_\dagger = P_\dagger^\omega] = \frac{1}{2^{b(1+|\mathcal{S}_{\mathrm{pfx}}|+\Lambda)}} = \frac{1}{2^{b|P_\dagger^\omega|}}.$$

Since the two samplings are independent, we have

$$\Pr[\tau_{\mathrm{id}} = \omega] = \Pr[P_\dagger = P_\dagger^\omega, P_\star = P_\star^\omega, K = K^\omega] = \frac{1}{2^{b|P_\dagger^\omega|}} \cdot \frac{1}{(2^b)_{q_p}} \cdot \frac{1}{2^\kappa}. \tag{2}$$

In the following lemma, we state a bound for the event BAD, which we prove in Section 5.3.

Lemma 1.

$$\begin{aligned}\Pr[\mathtt{BAD}] \leq\ & \frac{2q_c + q_p}{2^\kappa} + \frac{q_p(2\sigma + 6\Lambda + q_c)}{2^b} \\ & + \frac{\sigma(2\sigma + 2q_c + 2\Lambda + 1) + \Lambda(\Lambda + 2q_c + 1) + q_c(q_c + 1)}{2^b} \\ & + \frac{\sigma\Lambda}{2^{\min\{b, r+\kappa\}}} + \frac{\mathsf{mcoll}(\Lambda, 2^c) q_c}{2^r} + \frac{2\mathsf{mcoll}(\Lambda, 2^r) q_p}{2^c}.\end{aligned}$$

Let ω be any good transcript. From Eqns. (1) and (2), we have

$$\frac{\Pr[\tau_{\mathrm{re}} = \omega]}{\Pr[\tau_{\mathrm{id}} = \omega]} \geq 1. \tag{3}$$

Plugging in Lemma 1 and Eqn. (3) in Proposition 2 completes the proof of Theorem 1. □

5.3 Proof of Lemma 1

To bound $\Pr[\mathtt{BAD}]$, we bound the six bad events separately. The result follows from the union bound. We first observe that lines 19 and 20 of Algorithm 3 are always executed together, so there is an exact bijective correspondence between each prefix added to $\mathcal{S}_{\mathrm{pfx}}$ and each freshly sampled $Y_j^i \| Z_j^i$ added to $\mathsf{range}(P_\dagger)$. For each $\eta \in \mathcal{S}_{\mathrm{pfx}}$, let $Y_\eta \| Z_\eta$ denote the corresponding value in $\mathsf{range}(P_\dagger)$. Further, when $1 \leq j \leq m_i - 1$, $X_j^i \| W_j^i = Y_\eta \| Z_\eta \oplus M_j^i$ for some $\eta \in \{0\} \cup \mathcal{S}_{\mathrm{pfx}}$, and when $j = m_i$, $X_j^i \| W_j^i = Y_\eta \| Z_\eta \oplus M_j^i \oplus 0^{b-\kappa} \| K$ for some $\eta \in \mathcal{S}_{\mathrm{pfx}}$; we'll denote this η as η_j^i in both cases.

Bounding Pr $[\mathtt{BAD1}]$**:** This happens when $\exists\, i, i' \in [q_c], j \in [0, m_i + l_i - 1], j' \in [1, m_{i'} - 1]$ such that either $i < i'$ or $i = i', j < j'$, and $X^i_j \| W^i_j = X^{i'}_{j'} \| W^{i'}_{j'}$ with $M^i_1 \| \ldots \| M^i_j \neq M^{i'}_1 \| \ldots \| M^{i'}_{j'}$. Then $X^{i'}_{j'} \| W^{i'}_{j'} = Y_{\eta^{i'}_{j'}} \| Z_{\eta^{i'}_{j'}} \oplus M^{i'}_{j'}$, where $Y_{\eta^{i'}_{j'}} \| Z_{\eta^{i'}_{j'}}$ is sampled uniformly at random from $\{0,1\}^b$. Now, the definition of $X^i_j \| W^i_j$ depends on the index j, so we consider the following sub-cases:

- $\underline{\mathtt{BAD1.1}\text{: } j = 0}$. In this case, $X^i_j \| W^i_j = K \| IV$. For fixed i', j', since $Y_{\eta^{i'}_{j'}} \| Z_{\eta^{i'}_{j'}}$ is sampled uniformly randomly, the probability of this event is bounded by $1/2^b$. Taking union bound over the choices of i', j,

$$\Pr[\mathtt{BAD1.1}] \leq \frac{\sigma}{2^b}.$$

- $\underline{\mathtt{BAD1.2}\text{: } j \in [1, m_i - 1]}$. In this case, $X^i_j \| W^i_j = Y^i_{j-1} \| Z^i_{j-1} \oplus M^i_j = Y_{\eta^i_j} \| Z_{\eta^i_j} \oplus M^i_j$. If $\eta^i_j = \eta^{i'}_{j'}$, then $M^i_j \neq M^{i'}_{j'}$ (since $M^i \neq M^{i'}$), and there is no collision. Otherwise, $Y_{\eta^i_j} \| Z_{\eta^i_j}$ and $Y_{\eta^{i'}_{j'}} \| Z_{\eta^{i'}_{j'}}$ are sampled independently, and the probability of this equality holding is $1/2^b$. Taking union bound over the choices of i, i', j, j', we have

$$\Pr[\mathtt{BAD1.2}] \leq \frac{\sigma^2}{2^b}.$$

- $\underline{\mathtt{BAD1.3}\text{: } j = m_i}$. In this case, $X^i_j \| W^i_j = Y^i_{m_i-1} \| Z^i_{m_i-1} \oplus M^i_{m_i} \oplus 0^{b-\kappa} \| K = Y_{\eta^i_{m_i}} \| Z_{\eta^i_{m_i}} \oplus M^i_{m_i} \oplus 0^{b-\kappa} \| K$. In addition to the accidental full-state collision event, which occurs with probability $q_c\sigma/2^b$, for some i, i', j', it is possible that $\eta^i_{m_i} = \eta^{i'}_{j'}$. But this implies $M^{i'}_{j'} = M^i_{m_i} \oplus 0^{b-\kappa} \| K$, and the probability of this event is $1/2^\kappa$. Summing over all choices of i, we have

$$\Pr[\mathtt{BAD1.3}] \leq \frac{q_c}{2^\kappa} + \frac{q_c\sigma}{2^b}.$$

- $\underline{\mathtt{BAD1.4}\text{: } j \in [m_i + 1, m_i + l_i - 1]}$. In this case,

$$X^i_j \| W^i_j = T^i_{L,j-m_i} \| \left(Z^i_{m_i-1} \oplus \lfloor M^i_{m_i} \oplus 0^{b-\kappa} \| K \rfloor_c \oplus \left(\bigoplus_{k=m_i+1}^{j} T^i_{R,k-m_i} \right) \right).$$

  Now, we fix i, i', j, j' and look at the outer and inner parts separately. The equality in the outer part holds with probability $1/2^r$ by the randomness of $Y^{i'}_{j'-1}$. To bound the probability of the equality in the inner part holding, note that we cannot use the randomness of $Z^{i'}_{j'-1}$ since it is possible that $Z^{i'}_{j'-1} = Z^i_{m_i-1}$. But we can still use the randomness of K (or of $\lfloor K \rfloor_c$ if $\kappa > c$), so the equality holds with a probability of at most $1/2^{\min\{c,\kappa\}}$. Now, looking at the joint probability and taking union bound over all i, i', j, j', we have

$$\Pr[\mathtt{BAD1.4}] \leq \frac{\sigma\Lambda}{2^{\min\{b, r+\kappa\}}}.$$

Bounding Pr $[\mathtt{BAD2}]$**:** $\exists\, i, i' \in [q_c], j \in [0, m_i + l_i - 1], j' \in [1, m_{i'} - 1]$ such that either $i < i'$ or $i = i', j < j'$, and $Y^i_j \| Z^i_j = Y^{i'}_{j'} \| Z^{i'}_{j'}$ with $M^i_1 \| \ldots \| M^i_j \neq M^{i'}_1 \| \ldots \| M^{i'}_{j'}$. This bad event is triggered when $Y^{i'}_{j'} \| Z^{i'}_{j'}$ is defined. $Y^i_j \| Z^i_j$ and $Y^{i'}_{j'} \| Z^{i'}_{j'}$ are sampled independently, and hence,

$$\Pr[\mathtt{BAD2}] \leq \frac{\sigma^2}{2^b}.$$

Bounding Pr [BAD3]: This bad event is triggered if $\exists\, i, i' \in [q_c], j \in [0, m_i + l_i - 1], j' \in [m_{i'}, m_{i'} + l_{i'} - 1]$ such that either $i < i'$ or $i = i', j < j'$, and $X^i_j \| W^i_j = X^{i'}_{j'} \| W^{i'}_{j'}$. In this analysis, we have eight sub-cases, depending on the indices j and j'.

- BAD3.1: $j = 0, j' = m_{i'}$. This happens when

$$K \| IV = Y_{\eta^{i'}_{j'}} \| Z_{\eta^{i'}_{j'}} \oplus M^{i'}_{m_{i'}} \oplus 0^{b-\kappa} \| K.$$

  By the randomness of $Y_{\eta^{i'}_{j'}} \| Z_{\eta^{i'}_{j'}}$ we have

$$\Pr[\mathtt{BAD3.1}] \le \frac{q_c}{2^b}.$$

- BAD3.2: $j = 0, j' \in [m_{i'} + 1, m_{i'} + l_{i'} - 1]$. This happens when

$$K \| IV = T^{i'}_{L, j' - m_{i'}} \| \left(Z^{i'}_{m_{i'}-1} \oplus \lfloor M^{i'}_{m_{i'}} \oplus 0^{b-\kappa} \| K \rfloor_c \oplus \left(\bigoplus_{k = m_{i'}+1}^{j'} T^{i'}_{R, k - m_{i'}} \right) \right).$$

  By the randomness of $T^{i'}_{L, j'-m_{i'}}$ and $T^{i'}_{R, j'-m_{i'}}$ we have

$$\Pr[\mathtt{BAD3.2}] \le \frac{\Lambda}{2^b}.$$

- BAD3.3: $j \in [1, m_i - 1], j' = m_{i'}$. This happens when

$$Y_{\eta^i_j} \| Z_{\eta^i_j} \oplus M^i_j = Y_{\eta^{i'}_{m_{i'}}} \| Z_{\eta^{i'}_{m_{i'}}} \oplus M^{i'}_{m_{i'}} \oplus 0^{b-\kappa} \| K.$$

  Reasoning like in BAD1.3, we have

$$\Pr[\mathtt{BAD3.3}] \le \frac{q_c}{2^\kappa} + \frac{q_c \sigma}{2^b}.$$

- BAD3.4: $j \in [1, m_i - 1], j' \in [m_{i'} + 1, m_{i'} + l_{i'} - 1]$. This happens when

$$Y_{\eta^i_j} \| Z_{\eta^i_j} \oplus M^i_j = T^{i'}_{L, j'-m_{i'}} \| \left(Z^{i'}_{m_{i'}-1} \oplus \lfloor M^{i'}_{m_{i'}} \oplus 0^{b-\kappa} \| K \rfloor_c \oplus \left(\bigoplus_{k = m_{i'}+1}^{j'} T^{i'}_{R, k - m_{i'}} \right) \right).$$

  By randomness of $T^{i'}_{L, j'-m_{i'}}$ and $T^{i'}_{R, j'-m_{i'}}$, and applying the union bound to all possible choices of i, i', j, j', we have

$$\Pr[\mathtt{BAD3.4}] \le \frac{\sigma \Lambda}{2^b}.$$

- BAD3.5: $j = m_i, j' = m_{i'}$. This happens when

$$Y_{\eta^i_{m_i}} \| Z_{\eta^i_{m_i}} \oplus M^i_{m_i} = Y_{\eta^{i'}_{m_{i'}}} \| Z_{\eta^{i'}_{m_{i'}}} \oplus M^{i'}_{m_{i'}}.$$

  If $\eta^i_{m_i} = \eta^{i'}_{m_{i'}}$, then $M^i_{m_i} \neq M^{i'}_{m_{i'}}$, and this event does not happen. Otherwise, $Y_{\eta^i_{m_i}} \| Z_{\eta^i_{m_i}}$ and $Y_{\eta^{i'}_{m_{i'}}} \| Z_{\eta^{i'}_{m_{i'}}}$ are sampled independently, and the probability of this equality holding is $1/2^b$. Taking union bound over the choices of i, i', we have

$$\Pr[\mathtt{BAD3.5}] \le \frac{q_c^2}{2^b}.$$

- $\underline{\texttt{BAD3.6}\text{: } j = m_i, j' \in [m_{i'}+1, m_{i'}+l_{i'}-1]}$. This happens when

$$\begin{aligned}
&Y_{\eta^i_{m_i}} \| Z_{\eta^i_{m_i}} \oplus M^i_{m_i} \oplus 0^{b-\kappa} \| K \\
&\qquad = T^{i'}_{L,j'-m_{i'}} \| \left(Z^{i'}_{m_{i'}-1} \oplus \lfloor M^{i'}_{m_{i'}} \oplus 0^{b-\kappa} \| K \rfloor_c \oplus \left(\bigoplus_{k=m_{i'}+1}^{j'} T^{i'}_{R,k-m_{i'}} \right) \right).
\end{aligned}$$

As in $\texttt{BAD3.4}$, by randomness of $T^{i'}_{L,j'-m_{i'}}$ and $T^{i'}_{R,j'-m_{i'}}$ and union bound, we have

$$\Pr[\texttt{BAD3.6}] \leq \frac{q_c \Lambda}{2^b}.$$

- $\underline{\texttt{BAD3.7}\text{: } j \in [m_i+1, m_i+l_i-1], j' = m_{i'}}$. This happens when

$$\begin{aligned}
&T^i_{L,j-m_i} \| \left(Z_{\eta^i_{m_i}} \oplus \lfloor M^i_{m_i} \oplus 0^{b-\kappa} \| K \rfloor_c \oplus \left(\bigoplus_{k=m_i+1}^{j} T^i_{R,k-m_i} \right) \right) \\
&\qquad\qquad = Y_{\eta^{i'}_{m_{i'}}} \| Z_{\eta^{i'}_{m_{i'}}} \oplus M^{i'}_{m_{i'}} \oplus 0^{b-\kappa} \| K.
\end{aligned}$$

We look at the outer and inner parts of this equality separately. For the outer part, we need

$$T^i_{L,j-m_i} = Y_{\eta^{i'}_{m_{i'}}} \oplus \lceil M^{i'}_{m_{i'}} \oplus 0^{b-\kappa} \| K \rceil_r,$$

which happens with probability $1/2^r$ by the randomness of $Y_{\eta^{i'}_{m_{i'}}}$. For the inner part, we need

$$Z_{\eta^i_{m_i}} \oplus \lfloor M^i_{m_i} \rfloor_c \oplus \left(\bigoplus_{k=m_i+1}^{j} T^i_{R,k-m_i} \right) = Z_{\eta^{i'}_{m_{i'}}} \oplus \lfloor M^{i'}_{m_{i'}} \rfloor_c.$$

Since all $T^i_{R,k-m_i}$ are already sampled, the adversary can set this equation to be true by setting $\eta^{i'}_{m_{i'}} = \eta^i_{m_i}$ and choosing $M^{i'}_{m_{i'}}$ appropriately. This limits the number of choices for i, j, i' to $\mathsf{mcoll}(\Lambda, 2^c) q_c$. Also taking into account the accidental full-state collisions, we have

$$\Pr[\texttt{BAD3.7}] \leq \frac{\mathsf{mcoll}(\Lambda, 2^c) q_c}{2^r} + \frac{q_c \Lambda}{2^b}.$$

- $\underline{\texttt{BAD3.8}\text{: } j \in [m_i+1, m_i+l_i-1], j' \in [m_{i'}+1, m_{i'}+l_{i'}-1]}$. This happens when

$$\begin{aligned}
&T^i_{L,j-m_i} \| \left(Z^i_{m_i-1} \oplus \lfloor M^i_{m_i} \oplus 0^{b-\kappa} \| K \rfloor_c \oplus \left(\bigoplus_{k=m_i+1}^{j} T^i_{R,k-m_i} \right) \right) \\
&\qquad = T^{i'}_{L,j'-m_{i'}} \| \left(Z^{i'}_{m_{i'}-1} \oplus \lfloor M^{i'}_{m_{i'}} \oplus 0^{b-\kappa} \| K \rfloor_c \oplus \left(\bigoplus_{k=m_{i'}+1}^{j'} T^{i'}_{R,k-m_{i'}} \right) \right).
\end{aligned}$$

As in $\texttt{BAD3.4}$, by randomness of $T^{i'}_{L,j'-m_{i'}}$ and $T^{i'}_{R,j'-m_{i'}}$ and union bound, we have

$$\Pr[\texttt{BAD3.8}] \leq \frac{\Lambda^2}{2^b}.$$

Bounding Pr [BAD4]: $\exists\, i, i' \in [q_c], j \in [0, m_i + l_i - 1], j' \in [m_{i'}, m_{i'} + l_{i'} - 1]$ such that either $i < i'$ or $i = i', j < j'$, and $Y_j^i \| Z_j^i = Y_{j'}^{i'} \| Z_{j'}^{i'}$. Since $j' \in [m_{i'}, m_{i'} + l_{i'} - 1]$,

$$Y_{j'}^{i'} \| Z_{j'}^{i'} = T_{L,j'-m_{i'}-1}^{i} \| \left(Z_{m_{i'}-1}^{i'} \oplus \lfloor M_{m_{i'}}^{i'} \oplus 0^{b-\kappa} \| K \rfloor_c \oplus \left(\bigoplus_{k=m_{i'}+1}^{j'} T_{R,k-m_{i'}}^{i'} \right) \right).$$

We again use the randomness of $T_{L,j'-m_{i'}}^{i'}$ and $T_{R,j'-m_{i'}}^{i'}$, and apply the union bound to all possible choices of i, i', j, j' to obtain

$$\Pr[\texttt{BAD4}] \leq \frac{\sigma\Lambda}{2^b}.$$

Bounding Pr [BAD5]: For BAD5 to be true, we must have $X_j^i \| W_j^i = U_k$ for some $i \in [q_c], j \in [0, m_i + l_i - 1], k \in [q_p]$. We again divide this into four sub-events depending on the index j.

- BAD5.1: $j = 0$. For all $i \in [q_c]$, $X_0^i \| W_0^i = K \| IV$. Since K is sampled uniformly at random after the query phase,
$$\Pr[\texttt{BAD5.1}] \leq \frac{q_p}{2^\kappa}.$$
- BAD5.2: $j \in [1, m_i - 1]$. We must have $U_k = Y_{j-1}^i \| Z_{j-1}^i \oplus M_j^i$. Since $Y_{j-1}^i \| Z_{j-1}^i$ is generated in the offline phase by sampling uniformly from $\{0,1\}^b$, we have
$$\Pr[\texttt{BAD5.2}] \leq \frac{q_p \sigma}{2^b}.$$
- BAD5.3: $j = m_i$. We must have $U_k = Y_{m_i-1}^i \| Z_{m_i-1}^i \oplus M_j^i \oplus 0^{b-\kappa} \| K$. Since $Y_{m_i-1}^i \| W_{m_i-1}^i$ is generated in the offline phase by sampling uniformly from $\{0,1\}^b$, we have
$$\Pr[\texttt{BAD5.2}] \leq \frac{q_p q_c}{2^b}.$$
- BAD5.4: $j \in [m_i + 1, m_i + l_i - 1]$. We must have
$$U_k = T_{L,j-m_i}^i \| \left(Z_{m_i-1}^i \oplus \lfloor M_{m_i}^i \oplus 0^{b-\kappa} \| K \rfloor_c \oplus \left(\bigoplus_{\alpha=m_i+1}^{j} T_{R,\alpha-m_i}^i \right) \right).$$
  - First, suppose the i-th construction query is made after the k-th primitive query. In this case, since $T_{L,j-m_i}^i, T_{R,j-m_i}^i$ are chosen uniformly at random, this event happens with probability at most $1/2^b$. Varying over all i, j, k, the total probability of this case is upper bounded by $\Lambda q_p / 2^b$.
  - Next, suppose $\mathsf{dir}_k = +$ and the i-th construction query is made before the k-th primitive query. For this primitive query, the adversary has control over $\lceil U_k \rceil_r$ and can set it to $T_{L,j}^i$. Hence, for fixed i, j, k, the probability of this event happening is bounded by at most $1/2^c$ by the randomness of $Z_{m_i-1}^i$. Now since $T_{L,x}^i$ are chosen uniformly at random, by Proposition 1, given any $\lceil U_k \rceil_r$ the expected number of (i, x) such that $T_{L,x}^i = \lceil U_k \rceil_r$ is bounded by $\mathsf{mcoll}(\Lambda, 2^r)$. Hence, the total probability of this case is upper bounded by $\mathsf{mcoll}(\Lambda, 2^r) q_p / 2^c$.
  - Finally, suppose $\mathsf{dir}_k = -$ and the i-th construction query is made before the k-th primitive query. In this case, since U_k is generated as the output of the permutation P^{-1}, given any i, j, the probability of this case is bounded by at most $1/(2^b - k) \leq 1/2^{b-1}$. Varying over all (i, j, k), the total probability of this case is bounded by at most $\Lambda q_p / 2^{b-1}$.

$$\Pr[\texttt{BAD5.4}] \leq \frac{3\Lambda q_p}{2^b} + \frac{\mathsf{mcoll}(\Lambda, 2^r) q_p}{2^c}.$$

Bounding Pr [BAD6]: For BAD6 to be true, we must have $Y_j^i \| Z_j^i = V_k$ for some $i \in [q_c], j \in [0, m_i + l_i - 1], k \in [q_p]$. We again divide this into two sub-events depending on the index j.

- BAD6.1: $j \in [0, m_i - 1]$. Since $Y_j^i \| Z_j^i$ is generated in the offline phase by sampling uniformly from $\{0,1\}^b$, we have

$$\Pr[\mathtt{BAD6.1}] \leq \frac{q_p \sigma}{2^b}.$$

- BAD6.2: $j \in [m_i, m_i + l_i - 1]$. We must have

$$V_k = T^i_{L,j-m_i} \| \left(Z^i_{m_i-1} \oplus \lfloor M^i_j \oplus 0^{b-\kappa} \| K \rfloor_c \oplus \left(\bigoplus_{\alpha=m_i+1}^{j} T^i_{R,\alpha-m_i} \right) \right).$$

The analysis follows the exact steps of the analysis of BAD5.4 with the direction of the permutation queries reversed, and we obtain

$$\Pr[\mathtt{BAD6.2}] \leq \frac{3\Lambda q_p}{2^b} + \frac{\mathsf{mcoll}(\Lambda, 2^r) q_p}{2^c}.$$

5.4 Multi-user Security

In the multi-user setting with μ users, the oracles behave in exactly the same way as in the single-user case, with the following modifications:

- In the real world, the oracle samples μ independent keys $K_1, \ldots, K_\mu \xleftarrow{\$} \{0,1\}^\kappa$ at the outset.
- In the ideal world, the offline computation phase, μ independent keys $K_1, \ldots, K_\mu \xleftarrow{\$} \{0,1\}^\kappa$ are sampled.
- If the i-th construction query is made by user u_i, then $X_0^i \| W_0^i := K_{u_i} \| IV_{u_i}$.
- $K_{u_i} \| IV_{u_i}$ is also added as a prefix to each message for the purpose of keeping track of the prefixes during sampling.

Let $\mathcal{IV}$ denote the set of all distinct IVs and ν^{IV} denote the number of users with the common initialization vector IV. Let $\mathcal{L} = \max_{IV \in \mathcal{IV}} \nu^{IV}$.

The analysis of good transcripts remains unchanged. Differences arise only in the treatment of bad transcripts involving user keys. More specifically, it affects the bad events BAD1.1 and BAD5.1, because now multiple keys can be targeted. In addition, we have the following bad event.

BAD0: There exist distinct users $u, u' \in [\mu]$ such that $K_{u'} \| IV_{u'} = K_u \| IV_u$. Then we have

$$\Pr[\mathtt{BAD0}] \leq \sum_{IV \in \mathcal{IV}} \frac{\nu^{IV}(\nu^{IV} - 1)}{2^{\kappa+1}} \leq \frac{\mathcal{L}\mu}{2^\kappa}.$$

We also reanalyze the affected bad events below.

- BAD1.1. Now there are μ choices for $K_u \| IV_u$, so

$$\Pr[\mathtt{BAD1.1}] \leq \frac{\mu \sigma}{2^b}.$$

- BAD5.1. Since IV needs to be fixed for the primitive query, there are at most $\mathcal{L}$ choices for K_u, so

$$\Pr[\mathtt{BAD5.1}] \leq \frac{\mathcal{L} q_p}{2^\kappa}.$$

Updating these terms in the single-user bound and adding the bound for BAD0 yields the bound in Theorem 2, thus completing the proof. □

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